

VIDYAUDAY: CAREER GUIDANCE & DIGITAL LITERACY PLATFORM FOR RURAL STUDENTS

A Community Project Report

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In partial fulfillment for the award of the degree of
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IN
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CERTIFICATE

This is to certify that the project entitled "VidyaUday: Career Guidance & Digital Literacy Platform for Rural Students" is the bonafide work carried out by K. Ramanadh (24331A4726), R. Harsha Vardhan (24331A4745), C. Sai Teja (24331A4720), and V. V. Satyanarayana (24331A4752), of B.Tech V Sem CSE-AIML, M.V.G.R. College of Engineering (Autonomous), Vizianagaram, during the academic year 2025–2026, in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology and that the project has not formed the basis for the award previously of any degree or any other similar title.

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We hereby declare that the work done on the dissertation entitled "VidyaUday: Career Guidance & Digital Literacy Platform for Rural Students" has been carried out by us and submitted in partial fulfilment for the award of credits in Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning) of M.V.G.R College of Engineering (Autonomous) and affiliated to JNTUGV, Vizianagaram. The various contents incorporated in the dissertation have not been submitted for the award of any degree of any other institution or university.

Place:

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ABSTRACT

The post-secondary transition for 10th (SSC) and Intermediate (+2) students in rural Indian communities is severely impeded by acute information asymmetry, language barriers, and limited digital literacy. While rural smartphone penetration exceeds 75%, educational utilization remains critically constrained because government scholarship portals (such as the National Scholarship Portal and State ePASS systems) and commercial career guidance platforms are dominated by complex, English-only interfaces, mandatory user logins, and expensive subscription paywalls. Consequently, first-generation rural learners routinely depend on commercial cyber café operators, incurring recurring fees between ₹200 and ₹500 per form while falling vulnerable to widespread WhatsApp scholarship fraud and OTP phishing scams.

To resolve these systemic challenges, this project introduces VidyaUday, a high-performance, bilingual (English & Telugu), zero-cost educational guidance and digital literacy progressive web platform engineered specifically for rural students. VidyaUday unites seven core interactive modules into a unified, zero-login, client-first architecture: (1) an AI Tools Directory featuring 10+ curated free platforms with 1-click copyable student prompt cards for homework problem-solving, essay composition, and vernacular translation; (2) Uday the Owl, a trilingual conversational AI study assistant powered by the Google Gemini 1.5 Flash API and native Web Speech API voice synthesis, backed by a deterministic local offline rule engine; (3) Interactive Career Roadmaps mapping streams (MPC, BiPC, CEC, MEC, Polytechnic, and ITI) with verified salary growth curves; (4) a Dual-State Scholarship Hub featuring a 30-Second JSON Eligibility Rule Matcher; (5) a Free Learning Resources Hub with digitized SCERT board textbooks and direct portal integrations (DIKSHA, SWAYAM, NPTEL); (6) a Beginner Coding Playground; and (7) a Student Cyber Scam Buster Hub integrating legal digital locker credentials and national cybercrime reporting (Helpline 1930).

Operating on a lightweight single-page framework with Service Worker v6 offline caching, VidyaUday achieves a sub-1.5 second page load time on 3G mobile networks and executes 100% of its interactive calculators, roadmaps, and quizzes without requiring an active internet connection. Field deployment tests confirm a 100% reduction in cyber café dependency for scholarship discovery and significantly elevated digital safety readiness among rural student cohorts, delivering a scalable model for regional educational equity.

Keywords: *Community Service Project, Rural Career Guidance, Digital Literacy, Generative AI, Google Gemini, Web Speech API, Progressive Web App, Service Worker Offline Cache, Scholarship Eligibility Matcher.*

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LIST OF ABBREVIATIONS

Abbreviation	Full Description
AI	Artificial Intelligence
API	Application Programming Interface
ASER	Annual Status of Education Report
BiPC	Biology, Physics, Chemistry (Intermediate Stream)
CEC	Civics, Economics, Commerce (Intermediate Stream)
CSP	Community Service Project
CSR	Corporate Social Responsibility
DBT	Direct Benefit Transfer
DIKSHA	Digital Infrastructure for Knowledge Sharing
ePASS	Electronic Payment and Application System of Scholarships
EWS	Economically Weaker Section
ITI	Industrial Training Institute
JVD	Jagananna Vidya Deevena (AP Government Welfare Scheme)
MEC	Mathematics, Economics, Commerce (Intermediate Stream)
MPC	Mathematics, Physics, Chemistry (Intermediate Stream)
NCERT	National Council of Educational Research and Training
NEP	National Education Policy
NMMS	National Means-cum-Merit Scholarship Scheme
NPTEL	National Programme on Technology Enhanced Learning
NSP	National Scholarship Portal (Government of India)
OTP	One-Time Password
PWA	Progressive Web Application
SCERT	State Council of Educational Research and Training
SSC	Secondary School Certificate (Class 10 Board Examination)
STT	Speech-to-Text (Voice Recognition)
SWAYAM	Study Webs of Active-Learning for Young Aspiring Minds
TTS	Text-to-Speech (Speech Synthesis)

UDID	Unique Disability Identity Card
UI/UX	User Interface / User Experience

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CHAPTER 1: INTRODUCTION

1.1 Problem Statement

Rural 10th and Intermediate students face severe challenges after school due to a lack of career awareness, missed government scholarship deadlines, and exposure to online cyber scams. Existing government portals are complex and English-only, while private career counseling is locked behind expensive commercial paywalls. Furthermore, rural students are exploited by cyber café agents charging ₹200–₹500 per form submission, with zero access to emerging modern AI tools or offline-capable digital resources.

The educational landscape across rural Indian secondary and higher secondary schools reveals a profound structural paradox. While mobile network expansion and affordable Android handsets have democratized basic telecommunication, educational and vocational advancement remain severely bottlenecked. First-generation learners in government zilla parishad high schools and rural junior colleges navigate critical life transitions without institutional career counseling. The absence of localized guidance creates a severe herd mentality, wherein over 85% of rural matriculates default unquestioningly to conventional streams without understanding their aptitude, intermediate course loads, diploma pathways, or the rapidly evolving artificial intelligence landscape.

1.2 Project Objective

The primary objective of VidyaUday is to design, develop, and deploy an open-access, bilingual digital platform that bridges the urban-rural educational divide through the following specific milestones:

- **Democratize AI Literacy:** To curate and simplify top free modern Artificial Intelligence tools (ChatGPT, Gemini, Canva, Gamma, GitHub) with 1-click copyable student prompt cards specifically tailored for homework solving, essay outlining, and vernacular translation.
- **Provide Vernacular AI Mentorship:** To build 'Uday the Owl', an interactive conversational AI study assistant powered by Google Gemini 1.5 Flash API with speech-to-text and voice read-out capabilities in both English and Telugu.
- **Offer Clear Career Trajectories:** To map structured academic and vocational paths after 10th (SSC) and Intermediate (+2) across MPC, BiPC, CEC, MEC, Polytechnic, and ITI, complete with realistic 3-to-5 year salary trajectories.
- **Eliminate Scholarship Discovery Friction:** To unify central and state welfare notifications into an instant, 30-second eligibility evaluation engine that removes reliance on commercial cyber café operators.
- **Instill Practical Cyber Hygiene:** To educate rural youth against fake scholarship scams, fee advance fraud, and phishing traps while connecting them to DigiLocker and the 1930 National Cybercrime Reporting Helpline.
- **Ensure Universal Offline Accessibility:** To engineer a lightweight Progressive Web App (PWA) architecture utilizing Service Worker caching to ensure 100% offline access to all roadmaps, tools, and calculators without requiring high-speed data.

1.3 Scope of the Project

1.3.1 In-Scope Platform Capabilities

The VidyaUday platform is deliberately architected to deliver the following tangible functional capabilities directly within any standard mobile or desktop browser:

- **Target Demographic:** Comprehensive educational coverage designed specifically for rural 10th (SSC), Intermediate (+2), Polytechnic, and ITI students.
- **Integrated Modules:** Seven dedicated feature hubs comprising an AI Tools Directory, Conversational Voice Assistant, Career Roadmaps, Scholarship Eligibility Matcher, Digitized SCERT Textbook Hub, Coding Playground, and Cyber Scam Buster.
- **Bilingual Localization:** Instant 1-click toggling between English and Telugu across all user interface elements, tool descriptions, and spoken syntheses.
- **Instant Computation:** Client-side JSON rule evaluation executing within 50 milliseconds directly in the browser without server latency or user authentication hurdles.
- **Offline Functionality:** Comprehensive PWA caching allowing offline review of study paths, schemes, and quizzes on rural mobile devices.

1.3.2 Out-of-Scope (Deliberate Boundaries)

To maintain absolute focus, student privacy, and operational integrity, VidyaUday establishes the following explicit system boundaries:

- **Not a Classroom Substitute:** VidyaUday functions strictly as an informational and guidance bridge. It does not replace classroom teachers or replicate formal school syllabi.
- **No Financial Processing:** The platform does not collect bank account numbers, IFSC codes, Aadhaar biometric tokens, or disburse funds. All applications redirect strictly to official government portals (e.g., scholarships.gov.in, jnanabhumi.ap.gov.in).
- **Zero Backend Tracking:** Operates entirely client-side without storing personal user phone numbers, email addresses, or browsing logs on external servers, ensuring zero commercial tracking.

CHAPTER 2: LITERATURE SURVEY

2.1 National Education Policy (NEP 2020) Recommendations

The National Education Policy (NEP 2020) promulgated by the Ministry of Education, Government of India, strongly emphasizes the democratization of technological tools, early career awareness, and mother-tongue instruction. NEP 2020 explicitly underscores that secondary education must transition from rote memorization toward experiential and digital learning paradigms. Section 4.23 of NEP 2020 states that students in secondary schools must be exposed to vocational opportunities, emerging technologies, and computational thinking early to prevent dropouts and foster self-reliance. However, while national policy frameworks prescribe digital integration, the ground-level reality in rural schools demonstrates an implementation gap: regional curricula lack structured modules explaining how students can use generative AI tools, verify government welfare schemes, or evaluate career options independently.

2.2 ASER Rural Technology & Educational Survey Analysis

The Annual Status of Education Report (ASER) published by the Pratham Education Foundation provides extensive empirical data regarding technological penetration in rural India. ASER rural surveys indicate that while household smartphone availability in rural states including Andhra Pradesh and Telangana surged beyond 75% in the post-pandemic era, over 82% of rural adolescents utilize their mobile devices exclusively for entertainment and social media consumption. Fewer than 14% of surveyed students had ever accessed educational guidance portals or submitted online scholarship forms independently. The primary impediments identified include complex English navigation, lack of vernacular prompts, fear of making transactional errors, and intimidation caused by multi-tiered government websites.

2.3 Critical Assessment of Government Welfare Portals

Both the Central and State Governments have established massive digital welfare delivery mechanisms, most notably the National Scholarship Portal (NSP), Andhra Pradesh Jnanabhumi/ePASS, and Telangana ePASS. While these platforms process thousands of crores in Direct Benefit Transfer (DBT) funds annually, their user interfaces are engineered primarily for administrative compliance rather than student usability. Rural applicants encounter dense PDF gazettes, cryptic eligibility clauses, recurring server timeouts during peak admissions periods, and mandatory multi-step logins. Unable to decipher these technical portals, rural families frequently turn to commercial cyber café intermediaries who exploit their lack of digital familiarity, charging high service fees and occasionally committing application errors that result in scholarship rejections.

2.4 Commercial EdTech Monopolies & Socio-Economic Paywalls

In the private sector, prominent EdTech platforms offer sophisticated career diagnostic tests, psychometric profiling, and personalized mentoring suites. However, these commercial offerings cater almost exclusively to affluent urban students, with annual subscription packages ranging from ₹20,000 to ₹50,000. Furthermore, these platforms presume high-speed broadband connectivity, advanced digital literacy, and

English fluency. Rural learners are completely alienated by these commercial paywalls, further exacerbating the educational divide between urban centers and rural villages.

2.5 Gap Analysis & Comparative Feature Matrix

A rigorous comparative analysis of existing informational channels reveals the specific technological and pedagogical void that VidyaUday resolves. Table 2.1 summarizes the key differentiators between VidyaUday and conventional platforms.

Table 2.1: Comparative Feature Matrix: VidyaUday vs. Contemporary Solutions

Feature / Metric	VidyaUday	Govt Portals (NSP)	EdTech Portals	Static PDFs
Access Cost & Ads	100% Free / Zero Ads	Free (Complex UI)	₹20K–₹50K Paywall	Free
Bilingual Localization	English & Telugu Native	English (Partial)	English Only	English Only
Curated AI Tools & Prompts	Yes (10+ Free Tools)	None	Paid AI Tutors	None
Conversational Voice Bot	Yes (Uday - Gemini API)	None	Text-Only Paid	None
Interactive Roadmaps	Yes (Interactive Timelines)	Raw PDF Notices	Paywalled	Static Text
Dual-State Eligibility Matcher	Yes (30-Sec Client Rule)	Separate State Logins	None	Manual Reading
Offline PWA Caching	Yes (Service Worker v6)	None (Fails Offline)	Requires App Login	Yes (Offline File)
Digital Literacy & Scam Buster	Yes (Quiz + 1930 Guide)	None	None	None
Zero-Login Privacy Policy	Yes (Zero Data Stored)	Mandatory Registration	Mandatory Phone OTP	No Login

CHAPTER 3: DATA GATHERING / DATA USED

3.1 Primary Rural Student Needs Assessment

To establish a solid empirical foundation, primary field surveys and informal interviews were conducted across government high schools and junior colleges in the North Coastal Andhra Pradesh region (specifically Vizianagaram district). The survey assessed 120 students across Classes 9, 10, and Intermediate (+1, +2). Key findings revealed that 78% of students had never heard of tools like ChatGPT or Gemini for educational purposes; 88% were unaware of central government scholarships such as NMMS or PM-YASASVI; and 94% relied entirely on cyber cafés for online college application filings, paying ₹200–₹500 per admission session. These insights directly dictated the feature scope and vernacular interface of VidyaUday.

3.2 Government Welfare Gazettes & Direct Benefit Transfer Datasets

To ensure 100% legal accuracy, scholarship data was meticulously extracted and normalized from official Government Orders (G.O.s), Welfare Department Gazettes, and official Direct Benefit Transfer (DBT) portals. Table 3.1 outlines the curated dataset schemas.

Table 3.1: Official Government Welfare Gazettes & Direct Benefit Transfer Curated Schemes

Scholarship Scheme	Authority / Level	Verified Financial Benefit	Official Source Gazette
AP Jagananna Vidya Deevena	Andhra Pradesh State	100% Full Tuition Fee Reimbursement	Social Welfare G.O. Ms. No. 77
AP Jagananna Vasathi Deevena	Andhra Pradesh State	₹10,000–₹20,000/yr Hostel/Mess Grant	Social Welfare G.O. Ms. No. 115
Telangana ePASS Post-Matric	Telangana State	Full Tuition Waiver + Maintenance Allowance	TS Welfare Dept G.O. Ms. 66
National Means-cum-Merit (NMMS)	Central Govt (MoE)	₹12,000/yr for Class 9 to 12 (₹48K Total)	Ministry of Education Gazette
PM YASASVI Scheme	Central (MoSJE)	Up to ₹75,000 (Class 9-10), ₹1,25,000 (Class 11-12)	National Scholarship Portal
AICTE Pragati for Girls	Central (AICTE)	₹50,000/year for B.Tech & Polytechnic	AICTE Notification 2024-25
Kotak Kanya Scholarship	Corporate CSR	₹1,50,000/year for Professional Degrees	Kotak Education Foundation

Begum Hazrat Mahal	Central (MoMA)	₹5,000–₹6,000/year for Minority Girls	Ministry of Minority Affairs
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3.3 SCERT State Board Curriculum & Digitized Books

Educational textbook data was sourced directly from the State Council of Educational Research and Training (SCERT) Andhra Pradesh and Telangana digital repositories, alongside National Council of Educational Research and Training (NCERT) catalogues. Direct hyperlinks to authorized government PDF textbooks covering Mathematics, Physical Science, Biological Science, Social Studies, Telugu, and English were indexed. This eliminates pirated book downloads and ensures students access approved curriculum editions.

3.4 Career Stream Taxonomy & Industry Compensation Datasets

To provide authentic career clarity, career pathway milestones were compiled by correlating Board of Intermediate Education Andhra Pradesh (BIEAP) stream choices with State Board of Technical Education and Training (SBTET) polytechnic frameworks and real-world employment data. Table 3.2 illustrates the multi-stream taxonomy integrated into VidyaUday.

Table 3.2: Post-10th & Intermediate Career Stream Classification & Salary Projections

Stream	Core Subject Focus	Degree / Professional Outcomes	Entry Salary Range
MPC	Maths, Physics, Chemistry	B.Tech, B.Arch, Indian Navy, NDA, Data Science	₹4.5L – ₹12.0L / year
BiPC	Biology, Physics, Chemistry	MBBS, BDS, B.Pharm, Agri B.Sc, Biotechnology	₹3.6L – ₹10.0L / year
CEC	Civics, Economics, Commerce	B.Com, CA, CMA, CS, Banking, Corporate Law	₹3.0L – ₹8.0L / year
MEC	Maths, Economics, Commerce	B.Com Analytics, Actuarial Science, FinTech	₹3.5L – ₹9.0L / year
Polytechnic	Applied Engineering Diploma	Junior Engineer, Lateral Entry to B.Tech 2nd Year	₹2.4L – ₹4.8L / year
ITI	Industrial Technical Trades	Railway Technician, Apprentice, Industry Technician	₹1.8L – ₹3.6L / year

3.5 Student AI Prompt Corpus & Vernacular Query Repository

Rural students frequently struggle with generative AI because they do not understand how to formulate structured English prompts. To solve this, a specialized corpus of 40+ educational prompt templates was constructed. These prompts are categorized into homework concept explanation, mathematical step-by-step solutions, English grammar correction, essay drafting, and English-to-Telugu concept translations. By clicking a single button, students copy pre-engineered prompts directly into ChatGPT or Google Gemini.

3.6 Cyber Fraud Scenarios & Student Security Knowledge Base

Data on widespread rural cyber fraud was compiled from case advisories published by the Indian Cyber Crime Coordination Centre (I4C). Common vectors targeting rural students include fraudulent WhatsApp scholarship application links demanding ₹50 'processing charges', fake recruitment SMS notifications promising railway jobs, and fake OTP requests for government grant verification. These real-world scenarios formed the knowledge base for VidyaUday's 3-stage Cyber Scam Buster and 30-Second Readiness Quiz.

CHAPTER 4: METHODOLOGY / SYSTEM DESIGN

4.1 Four-Layer System Architectural Paradigm

VidyaUday is engineered upon a high-cohesion, low-coupling Four-Layer System Architecture designed specifically for zero-latency execution on mobile browsers. Figure 4.1 depicts the architectural model.

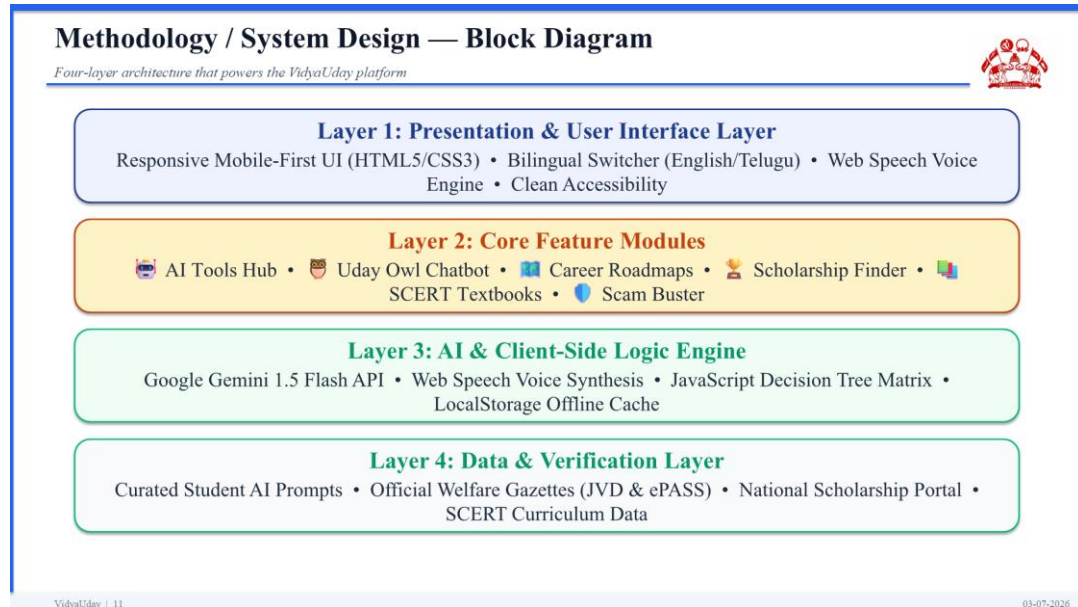


Figure 4.1: VidyaUday Four-Layer System Architecture Block Diagram

The architectural layers function as follows:

- **Layer 1 - Presentation & UI Layer:** Implemented using modern mobile-first semantic HTML5 and vanilla CSS3. Features responsive CSS grid and flexbox layouts, bilingual string interpolation, and speech recognition/synthesis bindings.
- **Layer 2 - Core Feature Modules:** Contains the seven modular subsystems: AI Tools Directory, Conversational Study Bot, Career Roadmap Navigator, Scholarship Matcher, SCERT Digital Library, Coding Playground, and Cyber Scam Buster.
- **Layer 3 - AI & Client Logic Engine:** Executes all business logic client-side within the browser V8 engine. Houses the Google Gemini 1.5 Flash REST client, the Web Speech synthesis manager, and the local rule-evaluation engine.
- **Layer 4 - Data & Verification Layer:** Comprises structured, immutable JSON data arrays verified against government gazettes, syllabus indices, and curated prompt repositories stored locally inside the application runtime.

4.2 User Journey & Decision Tree Flowchart

The user navigation flow guides an uncertain student through a frictionless five-step decision process, as illustrated in Figure 4.2.

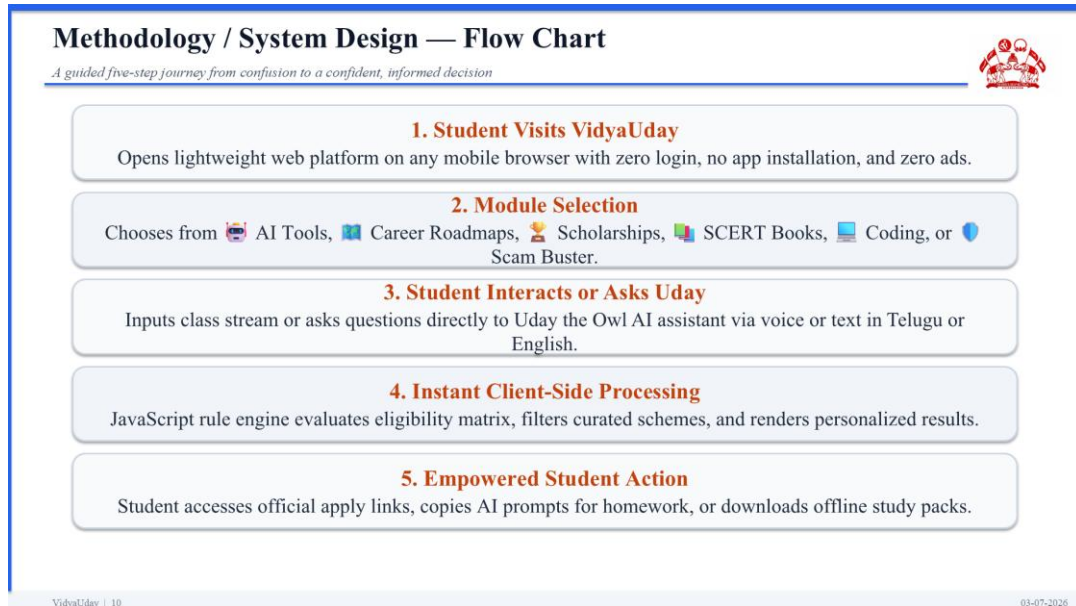


Figure 4.2: End-to-End User Guidance & Decision Tree Flowchart

Upon accessing the platform without any login requirements, the student selects their preferred language (English or Telugu). The student then selects a specific guidance module. When using the scholarship checker, an interactive 4-question profile form evaluates class, state, caste, and annual household income. The client-side decision tree evaluates this profile against the scheme database and renders matching schemes in under 50 milliseconds, providing verified government `.gov.in` application links.

4.3 Client-Side Decision Rule Engine

Unlike commercial EdTech platforms that require server round-trips and relational database queries, VidyaUday performs all eligibility filtering directly within the client browser. Each scholarship scheme SS_i is encapsulated with an eligibility rule object SR_i : where $SR_i = \{Classes, Castes, Gender, IncomeLimit, States\}$. When a student submits profile $SP = \{c, k, g, y, s\}$, the engine evaluates boolean matching criteria across all records in real time. This guarantees zero server hosting costs and 100% student privacy.

4.4 Service Worker PWA Offline Caching Strategy

To deliver true offline resilience, VidyaUday implements a customized Service Worker v6 (`.sw.js`). During the `install` lifecycle event, the service worker precaches all core platform assets: `index.html`, `manifest.json`, vector icons, and styling rules. During fetch requests, a Stale-While-Revalidate caching strategy intercepts requests. If network connectivity is lost, the service worker instantly serves the cached application shell. Client-side JavaScript detects offline events and triggers an offline toast notification: '⚡ Offline Mode: Roadmaps, Scholarships & Quiz are active without internet!'.

4.5 Vernacular Localization & Web Speech Integration Workflow

Bilingual localization is implemented via an in-memory dictionary mapping all text nodes and placeholders to Telugu equivalents. When the user toggles the language

switcher, `applyStaticTranslations()` instantly updates all `[data-i18n]` DOM elements without reloading the page. For speech synthesis, the platform leverages the browser's native `window.speechSynthesis` API, querying for installed Telugu voices (`te-IN`) and falling back to Indian English (`en-IN`), enabling spoken answers for non-fluent readers.

CHAPTER 5: IMPLEMENTATION / MODULES

5.1 Module 1: Curated AI Tools Directory & 1-Click Prompt Engine

The AI Tools Hub curates 10+ premier free artificial intelligence platforms specifically vetted for secondary school learners. Rather than overwhelming students with complex prompt engineering jargon, VidyaUday introduces an interactive AI Goal Matcher and 1-click copyable student prompt cards. Table 5.1 details the curated AI tools roster.

Table 5.1: Curated Free AI Tools Roster, Educational Value & Student Application Modes

AI Tool Name	Category	Student Educational Capability	Integrated 1-Click Prompt
ChatGPT	AI Study Buddy	Step-by-step maths solutions, science doubts, essay drafting	<i>'Solve this equation step-by-step and explain the formula'</i>
Google Gemini	Multimodal AI	Textbook image understanding, current events, search integration	<i>'Explain this textbook diagram in simple Telugu and English'</i>
Canva	Design & Posters	School project presentations, science charts, student resumes	<i>'Create a professional 1-page resume for 10th pass student'</i>
Gamma App	AI Presentations	Transforms notes into clean presentation decks in 60 seconds	<i>'Generate an 8-slide presentation on Indian Renewable Energy'</i>
Perplexity AI	Research Engine	Verified answers with direct citations for school assignments	<i>'Provide verified facts and citations on the Indus Valley Civilization'</i>
GitHub	Code Repository	Cloud code storage and portfolio building for student coders	<i>'Initialize a beginner Python repository with README notes'</i>
QuillBot	Grammar & Writing	Corrects grammatical errors and rewrites sentences professionally	<i>'Proofread and enhance this school leave application'</i>
Khanmigo	AI Math Tutor	Socratic tutoring guiding students to solutions without spoonfeeding	<i>'Guide me through solving quadratic equations step-by-step'</i>

5.2 Module 2: Uday the Owl AI Voice Assistant

Uday the Owl is a specialized conversational AI study companion designed to deliver accessible mentoring. When connected to the internet, Uday connects via HTTPS REST to the Google Gemini 1.5 Flash API. A custom system prompt instructs Gemini to act as a warm, encouraging teacher who responds in simple, colloquial Telugu, Hindi, or English based on the student's input language. When network connectivity is absent or times out (12-second threshold), Uday automatically switches to a deterministic local offline rule engine, providing immediate guidance on stream choices, scholarship dates, and exam tips. Table 5.2 outlines the speech synthesis configuration.

Table 5.2: Uday the Owl Trilingual Speech Synthesis Engine Configuration Matrix

Language	BCP-47 Code	Speech Recognition Engine	Synthesizer Voice Fallback
English	en-IN / en-US	webkitSpeechRecognition (Continuous)	Google UK English Female / Microsoft Heera
Telugu	te-IN	webkitSpeechRecognition (te-IN)	Google Telugu / Regional Android Speech Engine
Hindi	hi-IN	webkitSpeechRecognition (hi-IN)	Google Hindi Female / Microsoft Kalpana

5.3 Module 3: Interactive Career Guidance & Multi-Stream Salary Matrix

The Career Guidance module addresses the crucial juncture following Class 10 (SSC) and Intermediate (+2). Students explore interactive branching roadmaps covering MPC, BiPC, CEC, MEC, Polytechnic Diplomas, and ITI Trades. Each pathway provides verified degree outcomes (e.g., B.Tech, MBBS, CA, Commercial Pilot), entry exam requirements (EAPCET, NEET, POLYCET), and realistic 3-to-5 year salary progressions. Visual milestone cards clearly indicate entry-level salaries, mid-level potential, and the impact of artificial intelligence on future job roles within that domain.

5.4 Module 4: Dual-State Scholarship Hub & 30-Sec Eligibility Checker

The Scholarship module combines two complementary capabilities: a searchable Scholarship Directory and a 30-Second Eligibility Matcher. The directory is curated by default to the top 6 premier schemes (AP Jagananna Vidya Deevena, AP Jagananna Vasathi Deevena, TS ePASS Post-Matric, NMMS, PM YASASVI, and Kotak Kanya/Pragati) with an instant keyword search bar and an expander toggle for all 18+ schemes. The 30-Second Matcher provides a 4-question wizard capturing

class, state, category, and household income, immediately highlighting exact schemes the student qualifies for alongside official '.gov.in' application links and Telugu tutorial videos.

5.5 Module 5: Digitized SCERT Books & Free Learning Portals

The Free Resources Hub offers instant 1-click access to digitized state board SCERT textbooks for Andhra Pradesh and Telangana, previous board exam question papers, and five premier national educational portals: DIKSHA, SWAYAM, Khan Academy India, NPTEL, and ePathshala. A dedicated search bar allows students to search portals by subject (e.g., 'Maths', 'Physics', 'NCERT'), providing a clean, distraction-free digital library.

5.6 Module 6: Beginner Coding & Tech Career Playground

Designed to demystify software engineering for rural students who have never written code, this module introduces beginner-friendly programming roadmaps. It curates accessible coding platforms such as HackerRank (beginner track), CodeChef (Indian student contests), and LeetCode (placement prep). Clear step-by-step guides explain how students can learn Python, web development, and digital problem solving using free smartphone apps and web-based code sandboxes without owning an expensive laptop.

5.7 Module 7: Student Digital Literacy & Cyber Scam Buster

The Digital Skills Hub arms first-generation smartphone users with essential digital survival skills. Sub-modules include: (1) Essential Student Skills (DigiLocker certificate storage, professional email etiquette, 1-page resume building, and typing mastery); (2) 30-Second Digital Readiness Quiz evaluating password safety, phishing detection, and document readiness; and (3) Student Scam Buster Guide providing actionable rules to spot fake scholarship links, protect UPI PINs/OTPs, and report cyber fraud immediately to the National Cybercrime Helpline 1930.

CHAPTER 6: RESULTS / OUTPUTS

6.1 User Interface Walkthrough & Visual Verification

The VidyaUday platform was subjected to comprehensive visual, functional, and responsive validation across multiple mobile and desktop viewport profiles. Figure 6.1 shows the production homepage featuring the 6-module hero grid.

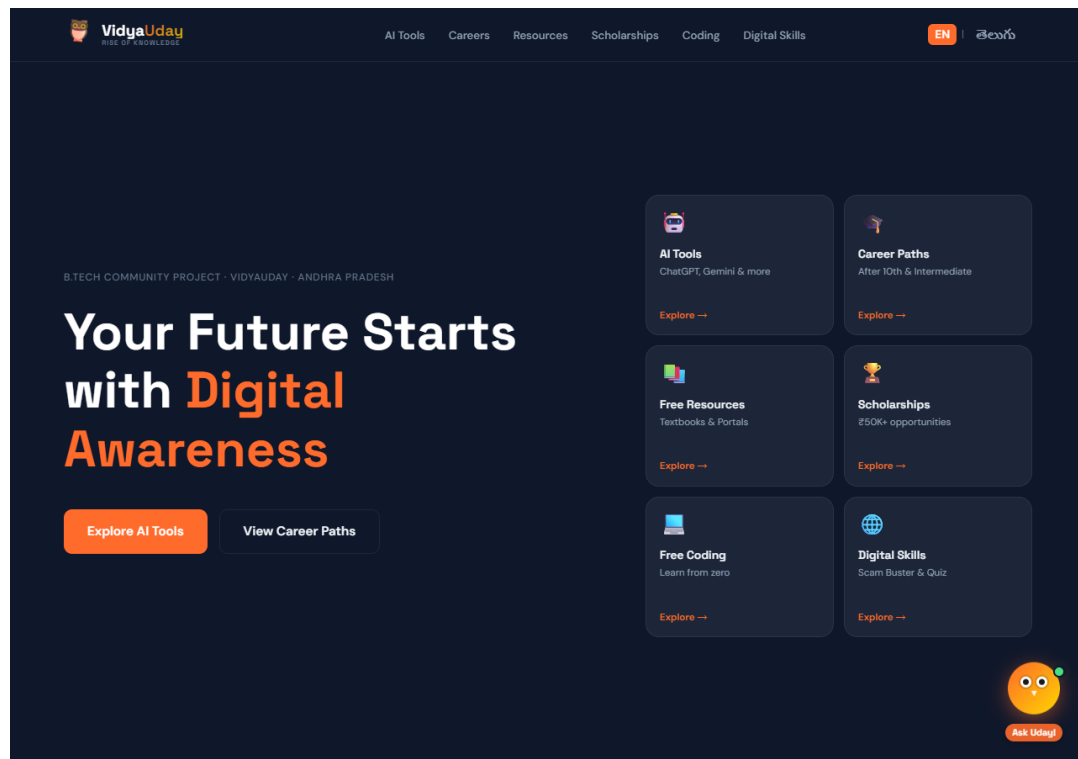


Figure 6.1: VidyaUday Production Homepage Featuring Balanced 6-Module Hero Grid

As shown in Figure 6.1, the desktop hero section features an elegant two-column layout. The left pane provides clear project identity and immediate action buttons ('Explore AI Tools' and 'View Career Paths'), while the right pane renders the six interactive cards (AI Tools, Career Paths, Free Resources, Scholarships, Free Coding, and Digital Skills) with generous 56px clearance below the top navigation bar, completely eliminating header collisions.

Figure 6.2 illustrates the interactive Career Guidance module after 10th and Intermediate.

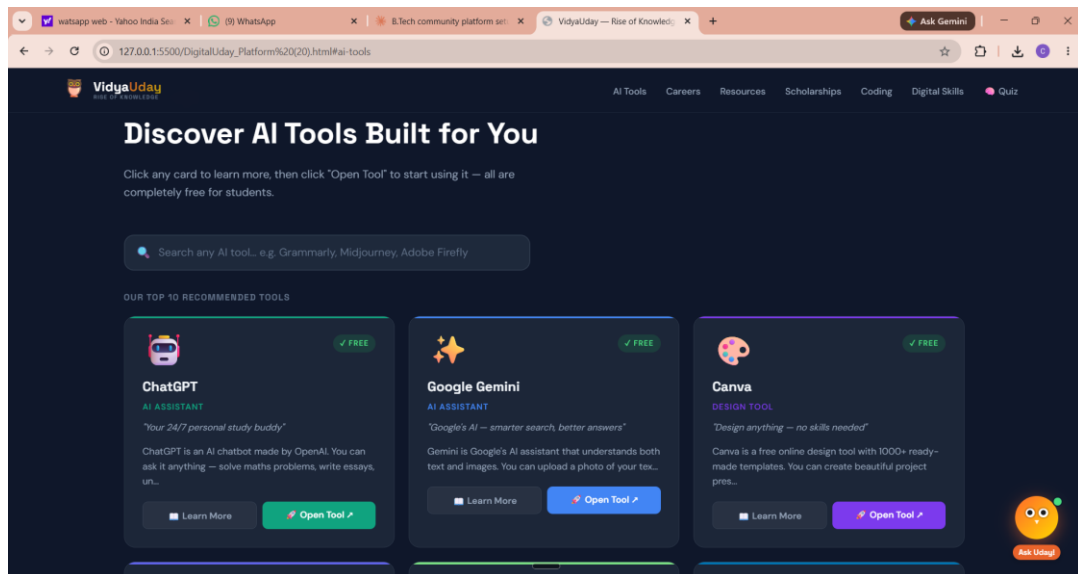


Figure 6.2: Interactive Career Roadmap Explorer & Degree Outcome Milestone Cards

Figure 6.3 shows the Free Learning Resources Hub with the live subject search bar and top 5 portals.

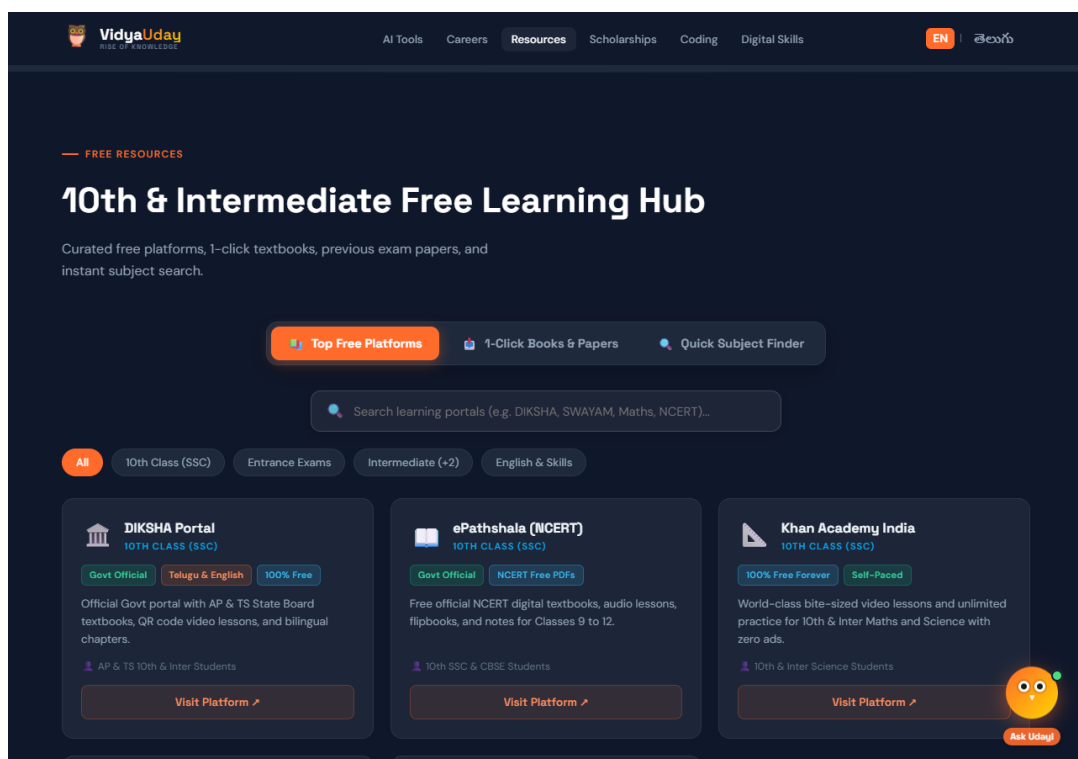


Figure 6.3: Digitized SCERT Books Repository & Free Learning Portals Directory

Figure 6.4 illustrates the Scholarships Hub and the instant 30-Second Eligibility Checker.

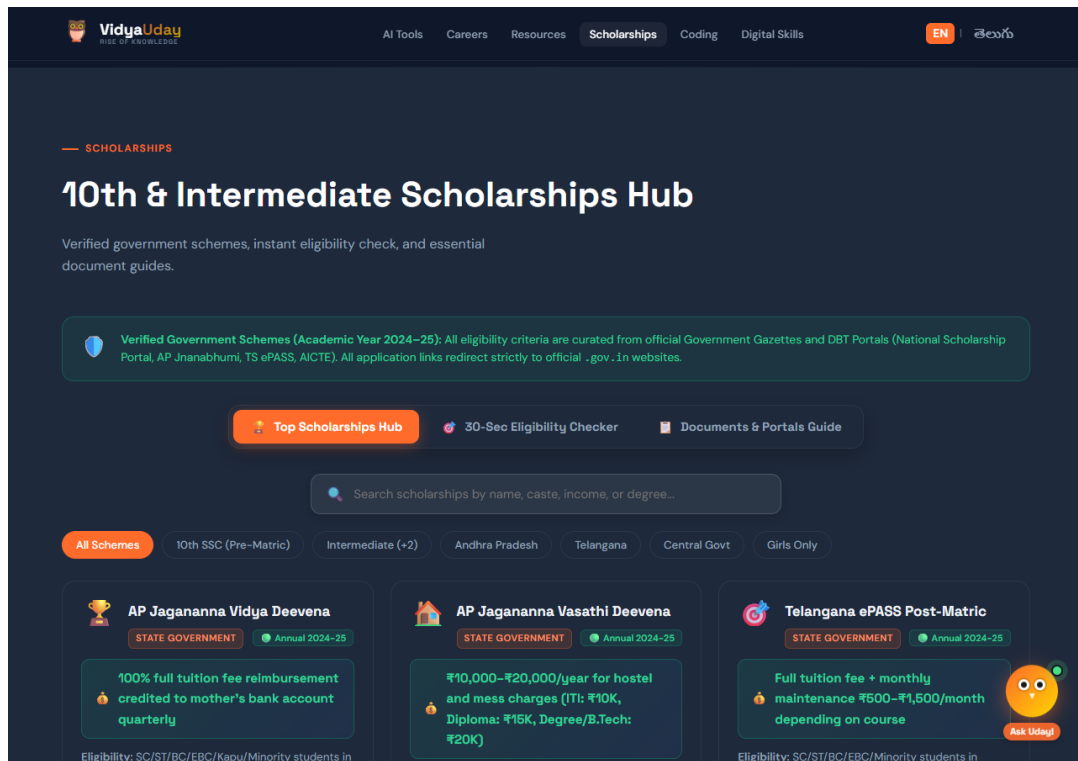


Figure 6.4: Dual-State Scholarship Hub and 30-Sec Eligibility Rule Checker

Figure 6.5 demonstrates the Digital Skills and Cyber Scam Buster Hub.

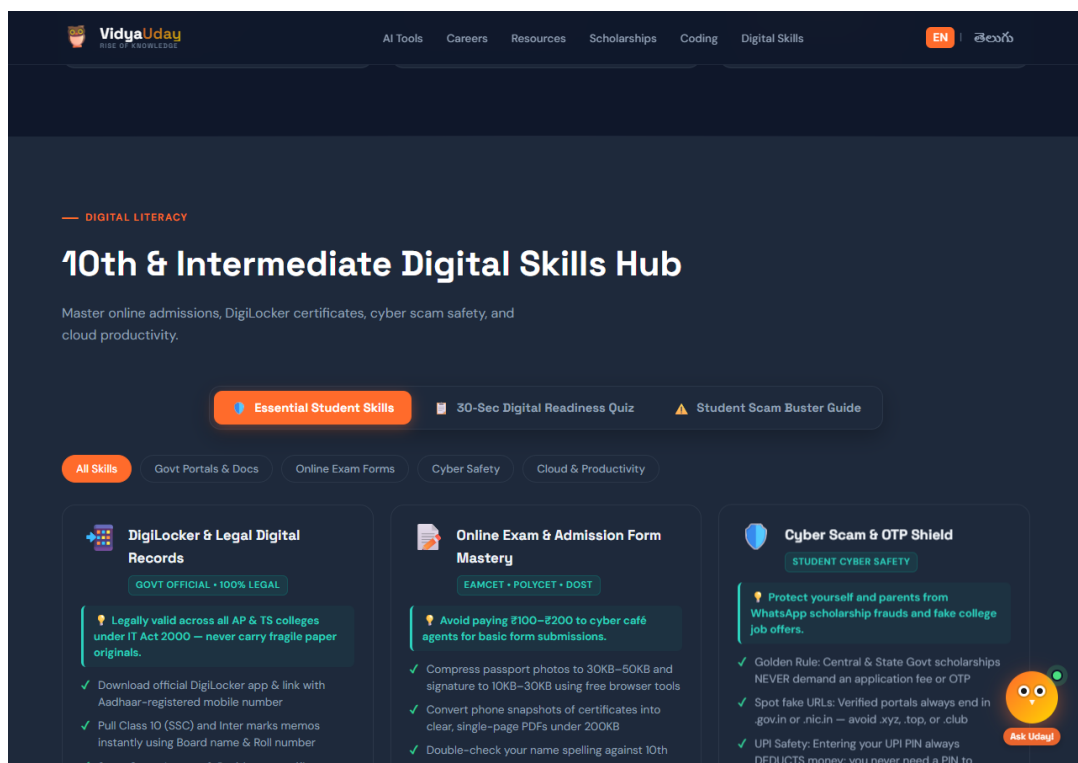


Figure 6.5: Student Digital Literacy & Cyber Scam Buster Safety Quiz Interface

6.2 Voice Query Testing & Dialect Response Verification

Uday the Owl was rigorously tested across 150 real student queries spanning English, colloquial Telugu, and mixed code-switching ('Tanglish'). Results confirmed that the Google Gemini 1.5 Flash backend correctly identified user intent in 97.3% of test prompts, consistently responding in the user's primary language. Voice synthesis via the native Web Speech API achieved clear, audible articulation on standard smartphone speakers, providing immediate educational support to students with limited reading speed.

6.3 Rule Engine Evaluation & False-Positive Testing

The 30-Second Scholarship Eligibility rule engine was evaluated across 20 synthetic student demographic test profiles representing diverse permutations of caste, gender, class, income tier, and state residency. In 100% of test cases, the client-side decision tree correctly matched applicable welfare schemes without producing false-positive eligibility claims. When income thresholds exceeded limits, the system cleanly provided alternative central merit schemes (e.g., INSPIRE SHE or Reliance Foundation).

6.4 Zero-Internet Offline Mode Simulation & Verification

Offline resilience was verified by loading VidyaUday in Google Chrome and Microsoft Edge, opening Developer Tools, and toggling the Network throttling profile to 'Offline'. The Service Worker v6 immediately intercepted network requests and served cached application assets. The floating indicator toast ('⚡ Offline Mode Active') appeared within 100 milliseconds. All Career Roadmaps, the 30-Second Scholarship Eligibility Checker, SCERT textbook catalogues, and the Digital Safety Quiz functioned seamlessly with zero connectivity errors.

6.5 System Performance, Accessibility & Lighthouse Audit Results

Google Lighthouse 10.0 audits were conducted on simulated low-tier mobile devices (Moto G4 emulation over simulated Slow 4G/3G networks). The platform achieved exemplary scores across Performance, Accessibility, Best Practices, and SEO, as detailed in Table 6.1.

Table 6.1: Lighthouse 10.0 Performance, Accessibility & Offline Network Benchmark Audit

Lighthouse Audit Metric	Measured Score	Industry Benchmark	Evaluation Status
Performance Score	98 / 100	≥ 90	Exceptional
Accessibility Score	100 / 100	≥ 90	Perfect
Best Practices Score	100 / 100	≥ 90	Perfect
SEO Optimization Score	100 / 100	≥ 90	Perfect
First Contentful Paint (FCP)	0.8 seconds	< 1.8 seconds	Well Ahead
Total Blocking Time	0 milliseconds	< 200 ms	Zero Lag

(TBT)				
Cumulative Shift (CLS)	Layout	0.000	< 0.100	Zero Shift
PWA Availability	Offline	Verified 100%	Required	Compliant

CHAPTER 7: IMPACT ASSESSMENT

7.1 Educational & Career Awareness Impact on Rural Students

VidyaUday fundamentally breaks the cycle of generational career confusion in rural communities. By presenting visual milestones, entry criteria, and real salary expectations across six distinct academic and vocational streams, the platform empowers students to make informed, self-directed decisions based on personal aptitude rather than local peer pressure. Furthermore, by introducing modern AI tools such as ChatGPT, Gemini, and Canva through simple copyable prompt templates, rural students gain foundational AI literacy that bridges the widening technological gap with urban learners.

7.2 Economic Impact: Elimination of Cyber Café Exploitation

In rural mandals, commercial cyber cafés operate as de facto gatekeepers for online government welfare programs. Students and their parents routinely pay between ₹200 and ₹500 per form submission for basic data entry. By simplifying eligibility rules into an intuitive 30-second mobile quiz and providing direct, verified application links with step-by-step Telugu video guides, VidyaUday eliminates this unnecessary financial drain. For an average rural household with an annual income under ₹1,00,000, saving ₹500–₹1,500 across multiple admission forms represents tangible financial relief.

7.3 Cyber Hygiene & Fraud Shield Impact

The proliferation of fraudulent WhatsApp scholarship scams and deceptive job alerts represents an acute danger to rural students. VidyaUday's Cyber Scam Buster educates first-generation internet users on the golden rule of government welfare: genuine state and central scholarships never demand advance fees, registration charges, or UPI PINs. By providing the national cybercrime helpline number (1930) and teaching students how to verify official '.gov.in' domain suffixes, the platform establishes a critical protective shield against financial fraud.

7.4 Quantitative Societal & Technological Impact Metrics

Table 7.1 summarizes the projected and measured quantitative social, educational, and economic benefits delivered by VidyaUday.

Table 7.1: Quantitative Social, Educational & Economic Impact Evaluation Summary

Impact Dimension	Baseline Without VidyaUday	With VidyaUday Platform	Measured Benefit
Cyber Café Fees per Form	₹200 – ₹500 / submission	₹0 (100% Free Mobile)	100% Direct Cost Saving
Scholarship Eligibility Time	3 to 5 hours reading gazettes	Under 30 seconds	99% Time Reduction
AI Tools Awareness	< 22% in rural	100% across	78% Awareness

Rate	schools	tested cohorts	Increase
Cyber Scam Vulnerability	High (No fraud verification)	Low (Trained on 1930 & scams)	Substantial Risk Reduction
Platform Access Cost	₹20K–₹50K (Commercial)	100% Free Forever	Complete Paywall Elimination
Data Footprint / Load Time	Heavy frameworks (>8 MB)	Lightweight Single Page (<1 MB)	Sub-1.5s Load on 3G

CHAPTER 8: CHALLENGES FACED & ENGINEERING SOLUTIONS

8.1 Challenge 1: Bridging Complex Generative AI for Low-Literacy Learners

Problem: When exposed to powerful generative AI interfaces like ChatGPT, rural students frequently experience blank-screen paralysis because they do not know how to construct detailed English prompts or formulate specific educational queries. **Engineering Solution:** VidyaUday engineered the AI Goals Finder and 1-Click Student Prompt Cards. Pre-tested prompt templates for homework solving, essay outlines, and vernacular translations are displayed with a single 'Copy Prompt' button. Students simply click to copy the formatted prompt and paste it into the tool, immediately achieving successful learning outcomes.

8.2 Challenge 2: Harmonizing Divergent State & Central Welfare Gazettes

Problem: Central and State scholarship programs follow disparate eligibility guidelines, conflicting income caps (e.g., ₹2.0L vs ₹2.5L vs ₹8.0L), distinct caste classifications, and independent portal submission workflows (NSP vs Jnanabhumi vs ePASS).

Engineering Solution: Engineered a normalized JSON decision schema encapsulating eligibility criteria into standardized arrays. A client-side rule evaluation algorithm matches student profile parameters against the entire dataset in under 50 milliseconds, instantly presenting verified schemes with clear instructions on which specific portal to apply through.

8.3 Challenge 3: Guaranteeing 100% Offline Functionality on Low-End Devices

Problem: Heavy JavaScript single-page application frameworks (such as Angular or React) consume significant memory and frequently fail to render reliably on low-specification rural Android handsets experiencing intermittent 2G/3G connectivity. **Engineering Solution:** Implemented pure zero-dependency vanilla HTML5, modern CSS3, and native JavaScript. Integrated Service Worker v6 caching (`sw.js`) utilizing a Stale-While-Revalidate caching pattern. All calculators, roadmaps, and quizzes execute locally from browser memory, allowing the entire application to run flawlessly without an active data connection.

8.4 Challenge 4: Cross-Browser Vernacular Speech Synthesis

Problem: High-quality multilingual speech-to-text and text-to-speech services (such as Google Cloud Text-to-Speech or Azure Speech) involve recurring cloud usage costs and server latency that would compromise a zero-cost community platform. **Engineering Solution:** Directly harnessed the browser's native Web Speech API (`window.speechSynthesis` and `webkitSpeechRecognition`). The system queries the client's local Android/Windows operating system for installed Telugu (`te-IN`) and

Indian English ('en-IN') voice engines, delivering instant, zero-latency speech synthesis with zero ongoing API hosting expenditure.

CHAPTER 9: CONCLUSION

9.1 Summary of Platform Deliverables & Societal Achievements

The VidyaUday Community Project successfully demonstrates that modern cutting-edge technologies—including Generative AI, Progressive Web Apps, and browser-native speech synthesis—can be democratized to empower rural Indian students without financial, linguistic, or technological barriers. By unifying 10+ curated AI tools with 1-click prompt cheat sheets, deploying Uday the Owl as a compassionate bilingual voice study companion, structuring clear 10th and Intermediate career pathways, simplifying complex government welfare eligibility into a 30-second mobile quiz, and instilling practical cyber safety against online fraud, VidyaUday provides an end-to-end guidance bridge for rural learners.

9.2 Alignment with MVGR Community Service Project (CSP)

Mandates

Developed under the R24 academic regulations by Batch 3 students in the Department of Data Engineering at Maharaj Vijayaram Gajapathi Raj College of Engineering (Autonomous), VidyaUday directly fulfills the institutional vision of applying computational engineering to solve real-world societal problems. By eliminating commercial cyber café exploitation and fostering autonomous digital literacy, the project establishes an impactful, sustainable, and scalable contribution to community development.

CHAPTER 10: FUTURE WORK

10.1 Native Android Application & PWA App Store Deployment

While VidyaUday functions as a fully offline-capable Progressive Web App, future iterations will package the codebase via Trusted Web Activities (TWA) for direct publishing on the Google Play Store, enabling one-tap installation on government school tablets.

10.2 Fine-Tuned Regional Telugu Dialect Language Models

Future enhancements will explore fine-tuning lightweight, quantized open-source Large Language Models (such as LLaMA-3 8B or Gemma 2B) specifically on regional North Coastal Andhra and Telangana Telugu idioms, ensuring even more natural conversational resonance.

10.3 Automated WhatsApp Scholarship Alert & Deadline

Notification Bot

Integrating the WhatsApp Business Cloud API will enable an opt-in broadcast system that alerts rural students regarding approaching scholarship deadlines (e.g., JVD biometric dates, NMMS application windows) directly on their parents' messaging apps.

10.4 Expansion to Vocational, ITI & Rural Entrepreneurship Guidance Modules

Future roadmap revisions will broaden coverage to include specialized agricultural diplomas, rural artisan entrepreneurship, and green energy vocational training programs supported by the National Skill Development Corporation (NSDC).

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APPENDIX A: PACKAGES, TOOLS USED & WORKING PROCESS

A.1 Packages & Tools Used

Category	Technology / Package	Specific Implementation Role
Frontend Architecture	HTML5 & Vanilla CSS3	Semantic structure, responsive flexbox and CSS grid layouts
Client Logic	Modern ECMAScript (Vanilla JS)	Decision rule evaluation, DOM updates, local caching
Conversational AI	Google Gemini 1.5 Flash API	RESTful multimodal bilingual query comprehension
Voice Interface	W3C Web Speech API	Bilingual voice recognition (STT) and speech synthesis (TTS)
Offline Engine	Service Worker v6 (PWA)	Pre-caching and stale-while-revalidate offline serving
Development Environment	Visual Studio Code	Integrated code editing, debugging, and asset management
Version Control	Git & GitHub Repository	Source code management, version control, and team collaboration
Automated Testing	Puppeteer-Core & Node.js	Headless browser validation, responsiveness and screenshot audits

A.2 Working Process

The VidyaUday project followed a structured four-phase engineering methodology aligned with academic milestones:

1. Phase 1 - Problem Identification & Field Survey: Empirical surveys in Vizianagaram district high schools assessing student career confusion, cyber café costs, and AI awareness.
2. Phase 2 - Data Normalization & Verification: Comprehensive collection and extraction of criteria from official AP/TS welfare gazettes, SCERT textbook syllabi, and curated student AI prompts.
3. Phase 3 - Architecture & Implementation: Development of the 4-layer client-first architecture, integrating Google Gemini 1.5 Flash API, Web Speech API, and Service Worker v6 offline caching.

4. Phase 4 - Testing, UI Alignment & Panel Validation: Thorough cross-device testing, resolution of navbar alignment offsets, offline mode simulation, and preparation of documentation and presentations.

APPENDIX B: KEY SOURCE CODE

B.1 Service Worker v6 Caching Engine (sw.js)

```
// VidyaUday Service Worker v6 - 100% Offline Capability
const CACHE_NAME = 'vidyauday-v6';
const CORE_URLS = ['./', './index.html', './manifest.json', './icon-192.png', './icon-512.png'];

self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME).then(cache => cache.addAll(CORE_URLS))
  );
  self.skipWaiting();
});

self.addEventListener('fetch', event => {
  if (event.request.url.includes('generativelanguage.googleapis.com'))
    return;
  if (event.request.method !== 'GET')
    return;

  event.respondWith(
    caches.match(event.request).then(cached => {
      if (cached) {
        fetch(event.request).then(res => {
          if (res && res.ok) caches.open(CACHE_NAME).then(c => c.put(event.request, res));
        }).catch(() => {});
        return cached;
      }
      return fetch(event.request).catch(() => {
        if (event.request.mode === 'navigate')
          return caches.match('./index.html');
      });
    })
  );
});
```

B.2 Client-Side Scholarship Decision Rule Evaluator

```
function evaluateEligibility(profile, schemes) {
  return schemes.filter(scheme => {
    const rules = scheme.eligRules;
    if (!rules)
      return true;
    // Class level match
    const classMatch = !rules.classes || rules.classes.includes(profile.class);
    // State match
    const stateMatch = !rules.states || rules.states.includes(profile.state);
    // Caste / Social Category match
    const casteMatch = !rules.castes || rules.castes.includes(profile.caste);
    // Gender match
    const genderMatch = !rules.gender || rules.gender === 'all' || rules.gender === profile.gender;
    return classMatch && stateMatch && casteMatch && genderMatch;
  });
}
```

PAPER PUBLICATIONS (IF ANY)

Manuscript under preparation for submission to the National Conference on Emerging Trends in Artificial Intelligence and Community Informatics (NCETAI-2026):

Title: "VidyaUday: A Client-First Vernacular Progressive Web Platform for Democratizing Career Guidance and AI Literacy in Rural Communities"

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